

Agentic AI for Public Administration

A multi-agent architecture and a controlled fault-injection evaluation for Romanian fiscal services

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Public services exist, yet stay fragmented

Romania has digitised fiscal services, but each portal has its own interface, vocabulary and procedural logic.

SPV · declarations & fiscal record

Ghişeu.ro · payments

RO e-Factura · national e-invoicing

A FREELANCER MUST

declare annual income

pay social and health contributions

register a rental contract

issue an electronic invoice

moving between disconnected systems, reconciling data no single platform brings together.

The real difficulty is coordination and interpretation, not arithmetic.

What the work sets out to do

- 1 Design** a layered multi-agent architecture: an entry agent routes each request to one of five domain agents over shared services
- 2 Instantiate** it for Romanian fiscal services on the UiPath Python SDK and LangGraph
- 3 Inject faults** at external service boundaries with a small set of chaos operators, deterministically replayable
- 4 Quantify & explain** how many faults propagate silently, in terms of the operator mix and response-schema structure
- 5 Estimate cost** of running such a system at institutional scale

If realistic faults hit the service boundaries, how many reach the citizen unnoticed, and what governs that number?

Standing on three bodies of work

Agentic orchestration

LangGraph models control as a typed state machine: explicit and auditable, unlike open-ended conversation. Decisive for a regulated service.

Retrieval (RAG)

Tax law is specific and frequently amended. RAG grounds answers in the Romanian Tax Code, kept separate from deterministic computation.

Chaos engineering

Introduce faults deliberately and observe. Recently reaching LLM agents, but usually in generic or synthetic testbeds.

THE RESEARCH GAP

No prior work brings an operational multi-agent architecture for a national fiscal setting together with a boundary-level fault study that explains *why* faults pass silently, in terms of schemas and predicates.

A layered multi-agent stack

Orchestration · Entry Agent *intent detection & confidence-gated routing*

Domain agents (5) *PFA · Property sale · Rental · Certificate · e-Factura*

Shared services *Document intake · RAG · Calculation · Payment*

Infrastructure (UiPath SDK) *model access · retrieval · storage · human review*

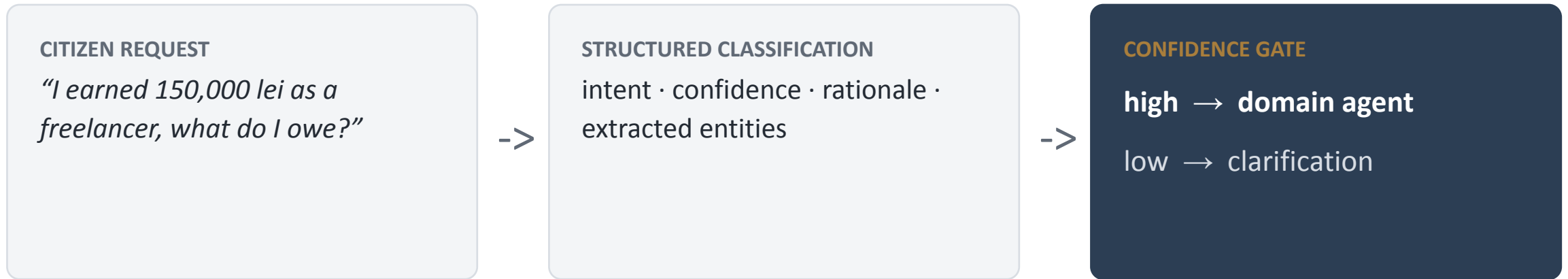
Mocked boundaries → **government systems** *SPV · Ghişeu.ro · RO e-Factura*
(schema-faithful mocks)

FAULT-INJECTION LAYER

Sits at the last seam, above the mocked boundaries and separate from the agents' logic.

Design principle: interpretation is done by the model, computation by deterministic code.

One entry, confidence-gated routing



ROUTING IN TWO STAGES

A deterministic mapping turns the classified intent into a graph node.

A confidence threshold diverts ambiguous requests to clarification rather than to a possibly wrong specialist.

In a fiscal setting, dispatching the wrong specialist could mean computing the wrong tax.

IMPLEMENTATION

Five workflows over four shared services

DOMAIN AGENTS

PFA / Freelancer Declarația Unică; CAS & CASS via calculation service

Property sale tax by holding period; payment via Ghișeul.ro

Rental income contract registration + flat tax on annual rent

Certificate certificat de atestare fiscală: type, docs, submission

e-Factura B2B/B2C invoicing; XML generation + status monitoring

SHARED SERVICES

Document intake

OCR + schema validation; below 80% to manual review

RAG

grounds explanations in the Romanian Tax Code

Calculation

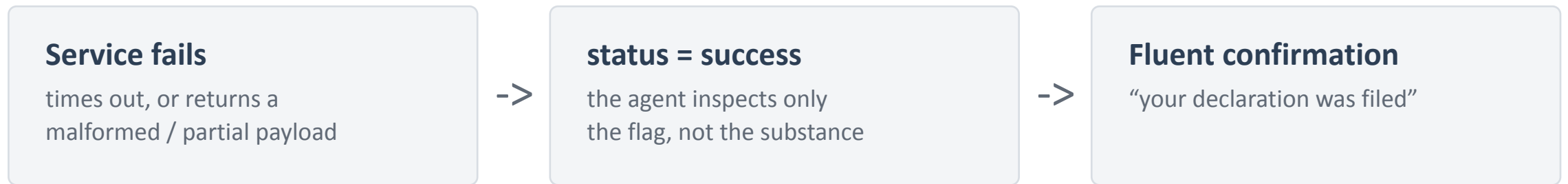
deterministic, decimal; rates read from configuration

Payment

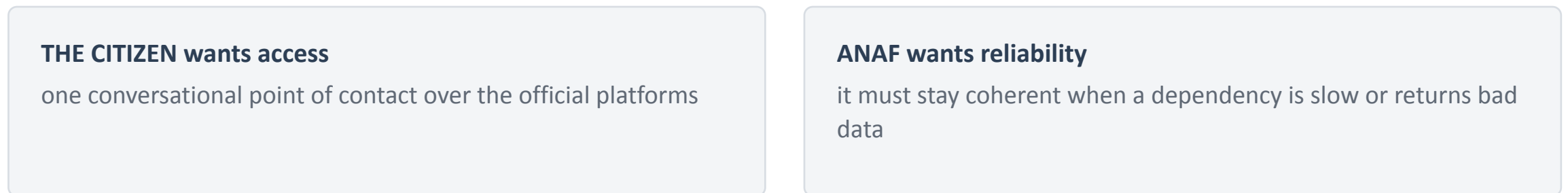
builds request, returns Ghișeul.ro redirect URL

The danger is a quiet one

An agent that trusts a service's **success flag** and looks no further will build a confident answer on top of a broken response.



The conversation reads as normal while a broken state slips through beneath it.



Perturb each boundary crossing

THE INJECTOR

A decorator wraps each mocked boundary. One Bernoulli trial per call decides if a fault fires.

seed s deterministic replay

prob. η per-call injection rate

Δt max bounds the delay

$\eta = 0$ recovers the nominal system, unchanged.

FOUR CHAOS OPERATORS

Timeout

raises an error, no payload returned

always caught

Delay

sleeps, then returns the payload intact

always silent

Partial

sets one random field to None

usually silent

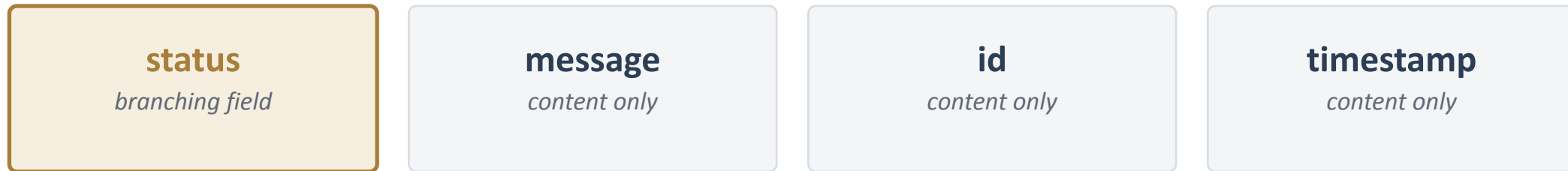
Corruption

replaces one field with a sentinel

usually silent

Why faults pass silently

Agent control flow usually depends on **one field**: a predicate like `status == "success"`.



A single-field fault that misses **status** leaves the predicate true. The agent takes the success branch anyway.

SILENT PASS-THROUGH IS A PROPERTY OF THE SCHEMA

$$P(\text{silent}) = 1 - 1/|F|$$

with $|F|$ fields, one of which gates control flow

4-field response object

0.75 silent

RESULTS

~63% silent, and it is predictable

SILENT PASS-THROUGH

~63%

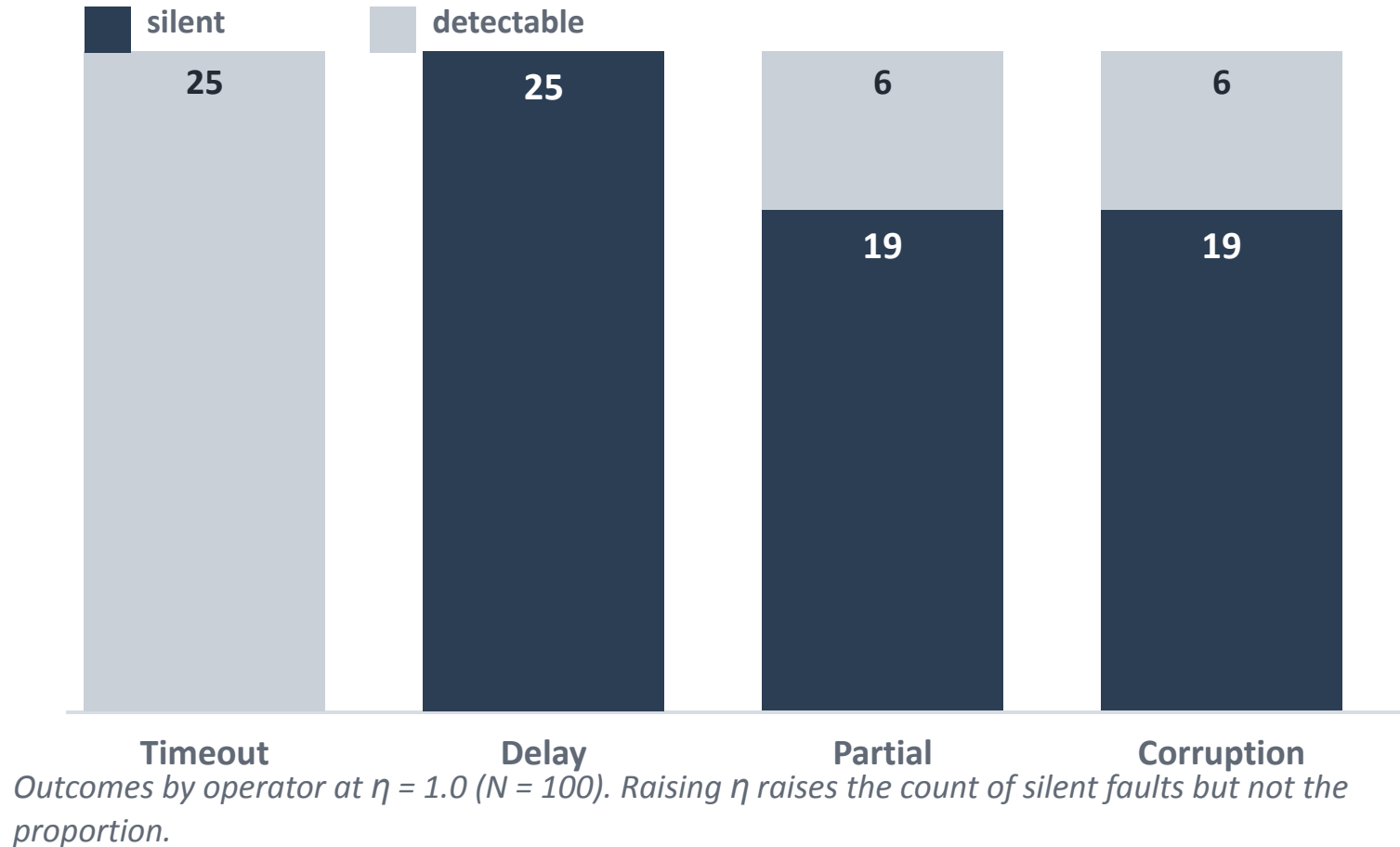
across every non-zero injection rate

MATCHES THEORY

$0.25 + 2 \times 0.25 \times 0.75$

= 0.625

*delay + partial + corruption; timeout
always caught*



Inference is not the barrier

PER SESSION

~5¢

~20k input + 2.1k output tokens (GPT-4o)

SMALL OFFICE

\$51

/ month
~50 sessions / day

LARGE AGENCY

\$5,060

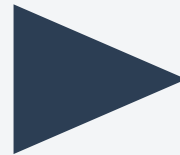
/ month
~5,000 sessions / day

Cost scales linearly and stays modest. Routing with a cheaper model, reserving the capable one for reasoning, pushes it lower still.

Whatever stands in the way of deployment is far more likely to be assurance than arithmetic.

DEMONSTRATION

A walkthrough of a citizen session



Insert the demo video here

freelancer asks what they owe · classify & route · compute CAS/CASS · guide submission

Interface coherence is not evidence of correctness

FINDINGS

- * A structural ~63% silent rate, matching a closed-form prediction
- * The cause is in response schemas and predicates, not incidental bugs
- * Inference cost is modest; assurance is the real barrier

WHAT COMES NEXT

Output contracts

validate responses before they reach the citizen

Trace-based assurance

structured traces, deterministic replay, governance

Beyond taxation

other public-service domains

Thank you

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